

Irrigation Measurement

Field guide to the AGHRI irrigation calculators

— MEASURE WATER. MANAGE EVERY DROP.

A portable companion to the AGHRI Lab irrigation calculators, with four field methods, equations, protocols, a field record sheet, unit conversions, and technical references. Use it to turn field observations into irrigation-ready numbers with conditions documented well enough to be defended later.

— WHAT'S INSIDE

01 Four measurements

Discharge, applied depth, uniformity, and root-zone deficit with formulas.

02 Field protocols

Short, repeatable steps and the field kit for each measurement.

03 Field record sheet

A printable page for site details, replicates, and conversions.

04 Technical references

The manuals and peer-reviewed sources behind the formulas.

— USE WITH CARE

Use this guide for field calculations and training. **Verify measurements, assumptions, system limits, and crop requirements before making operational decisions.** Companion calculators:

osuagrohydrology.com/pages/irrigation-calculators.html

COMPANION TOOLS

osuagrohydrology.com
Irrigation calculators and the
Oregon scheduling tool directory

AGHRI LAB

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Four measurements. Four decisions.

Use the formulas with consistent units. Repeat observations and record operating conditions.

TIMED-VOLUME DISCHARGE

$$Q = V \div t$$

Q is discharge; V is collected volume; t is collection time. For US gallons collected over seconds: $Q \text{ (gpm)} = V \times 60 \div t$.

FIELD NOTE Use a calibrated container and repeat the test at least three times.

APPLIED IRRIGATION DEPTH

$$d = V \div A$$

US units: gross depth (in) = applied gallons $\div$ [area (ac) $\times$ 27,154]. Metric: gross depth (mm) = volume (m^3) $\div$ [area (ha) $\times$ 10]. Net depth = gross depth $\times$ application efficiency.

FIELD NOTE Use the actual wetted area and document the efficiency value.

CATCH-CAN UNIFORMITY

$$DU_{lq} = x_{lq} \div x_{avg}$$

Low-quarter distribution uniformity is the mean of the lowest 25% of catches divided by the overall mean. Christiansen CU = $100 \times [1 - \Sigma |x_i - x_{avg}| \div (n \times x_{avg})]$.

FIELD NOTE Use matched, level cans and record wind, pressure, test duration, and anomalies.

ROOT-ZONE SOIL-WATER DEFICIT

$$SWD = (\theta_{fc} - \theta_v) \times Z_r$$

SWD is net soil-water deficit; θ_{fc} is volumetric water content at field capacity; θ_v is current volumetric water content; Z_r is active root depth. Gross refill = $SWD \div$ application efficiency.

FIELD NOTE Texture presets are screening values only. Use site-calibrated soil limits when possible.

Interpretation boundary: A calculator result is only as representative as the sampling design, input accuracy, and field conditions observed during the test.

Field protocols

A short, repeatable method beats a precise-looking number with undocumented conditions.

01 Timed volume

FIELD KIT

Calibrated container, stopwatch, field sheet

1. Choose a safe collection point with steady flow.
2. Time a complete fill; avoid estimating partial seconds.
3. Repeat at least three times and average the discharge.
4. Record pressure or operating configuration when relevant.

02 Flow × runtime

FIELD KIT

Flow measurement, runtime record, wetted area

1. Confirm flow represents the complete operating set.
2. Use the actual wetted area, not the parcel area.
3. Calculate both gross and net applied depth.
4. Document the efficiency assumption and its source.

03 Catch-can test

FIELD KIT

Matched cans, ruler, flags, pressure and wind readings

1. Place identical cans on a regular representative grid.
2. Level cans and avoid crop-canopy interception.
3. Run long enough to obtain measurable depths.
4. Flag spills, plugged nozzles, pressure changes, and wind shifts.

04 Root-zone deficit

FIELD KIT

Calibrated sensor, root-depth estimate, soil limits

1. Sample more than one location and depth.
2. Use measured field capacity and wilting point when available.
3. Use the active crop root depth, not total soil depth.
4. Check the refill against infiltration rate and system capacity.

Field record sheet

Document the observations that make the calculation defensible. Print this page.

SITE / FIELD

DATE / TIME

SYSTEM / SET

OPERATOR

CROP / STAGE

SOIL / ROOT DEPTH

PRESSURE

WIND / WEATHER

— OBSERVATIONS AND REPLICATES

REPLICATE	MEASURED VALUE	UNIT	CONDITION / NOTE
1			
2			
3			
4			
5			
6			

— QUICK CONVERSIONS

1 acre-inch
27,154 US gal

1 cubic foot per second
448.8 US gpm

1 US gpm
0.06309 L/s

1 inch
25.4 mm

1 acre
0.404686 ha

1 US gallon
3.78541 L

— RESULT / DECISION

Technical references

The primary manuals, standards publications, and peer-reviewed sources behind the formulas.

- 01** Allen, R. G., Pereira, L. S., Raes, D., & Smith, M. (1998). Crop evapotranspiration: Guidelines for computing crop water requirements. FAO Irrigation and Drainage Paper 56. Food and Agriculture Organization of the United Nations. **FAO publication**
- 02** Burt, C. M., Clemmens, A. J., Strelkoff, T. S., Solomon, K. H., Bliesner, R. D., Hardy, L. A., Howell, T. A., & Eisenhauer, D. E. (1997). Irrigation performance measures: Efficiency and uniformity. Journal of Irrigation and Drainage Engineering, 123(6), 423-442. **Journal article**
- 03** Christiansen, J. E. (1942). Irrigation by sprinkling. California Agricultural Experiment Station Bulletin 670, University of California, Berkeley. **Digital copy**
- 04** Merriam, J. L., & Keller, J. (1978). Farm irrigation system evaluation: A guide for management. Utah State University. **USU record**
- 05** Oregon State University Extension Service. Efficient irrigation of perennial forages with wheel lines in the Pacific Northwest (EM 9399). Authors: I. McGregor, M. I. Zamora Re, G. B. Jones, and T. Peters. **OSU Extension**
- 06** U.S. Department of Agriculture, Natural Resources Conservation Service. (1997). National Engineering Handbook, Part 623, Chapter 9: Water Measurement. **USDA NRCS PDF**
- 07** U.S. Department of Agriculture, Natural Resources Conservation Service. (1997). National Engineering Handbook, Part 652: Irrigation Guide. **USDA NRCS PDF**
- 08** U.S. Department of the Interior, Bureau of Reclamation. (2001). Water Measurement Manual (3rd ed., revised reprint). **Bureau of Reclamation PDF**

— SOURCE NOTE

Reference links were verified against institutional, government, publisher, or university repository pages in August 2026. Soil texture presets in the interactive calculators are approximate starting values and are not substitutes for field-calibrated limits.

Interactive calculators: osuagrohydrology.com/pages/irrigation-calculators.html